

Lethal levels of pH for brook trout, *Salvelinus fontinalis* (Mitchill)

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Fingerling brook trout, *Salvelinus fontinalis* (Mitchill), were subjected to acute and chronic pH stress for maximum periods of 10 000 min in a series of dilutions of sulfuric acid or sodium hydroxide at 10 and 20°C after acclimation at 15°C and pH 6.8. Lethal limits of pH were estimated to be pH 3.5 and 9.8 at both temperatures. There were no observable differences in the resistance times at lethal pH stresses between test temperatures. The order of death was not significant in relation to length or weight of the fish.

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Des ombles de fontaine juvéniles, *Salvelinus fontinalis* (Mitchill), ont été soumises à des stress aigus et chroniques de pH pour des périodes d'une durée maximale de 10 000 min dans une série de dilutions d'acide sulfurique ou de soude à 10 et 20°C, après acclimation à 15°C, à un pH de 6.8. Les limites létales de pH sont estimées à 3.5 et 9.8 aux deux températures. Il n'y a pas de différences appréciables des durées de résistance au pH létales entre les deux températures expérimentales. Il n'y a pas de relation significative entre la séquence des morts et la longueur ou le poids des poissons. [Traduit par le journal]

Introduction

In consideration of the relatively long interval during which pH has been measured routinely in limnological surveys, it has been associated rather imperfectly in the physiological performance of fishes. Some aspects of activity responses have been reported by Wells (1915), Brown and Jewell (1926), Breder (1927), Gutsell (1929), Creaser (1930), and Jones (1948). Reports on experimental studies of lower and upper lethal limits of pH are considerably less than adequate. The lethal limits of ambient pH have been determined for only 11 species of fish including two salmonids: brown trout, *Salmo trutta*, and rainbow trout, *Salmo gairdneri* (Dahl 1927; Bandt 1936; Schiemenz 1937; Jones 1939; Huet 1941; Sanborn 1945; Eicher 1946; Stiemke and Eckenfelder 1947; Vivier 1954; Trama 1954; Meinck *et al.* 1956; McKee and Wolf 1963; Lloyd and Jordan 1964; Jordan and Lloyd 1964; Beamish 1972). Approximate lethal levels of pH ranged from 3.7 to 5.4 in acidic waters and from pH 8.7 to 11.1 in alkaline waters for these fishes.

In the present study, as a preliminary consideration of structural alterations resulting from exposure to pH at and near the lethal limits, samples of brook trout, *Salvelinus fontinalis*

(Mitchill), were exposed to a series of pH levels below and above 6.8 to determine their tolerances at two temperatures.

Materials and Methods

Hatchery-reared fingerling brook trout, *Salvelinus fontinalis* (Mitchill), were obtained from the Department of Environment, Coldbrook Fish Culture Station, Coldbrook, Kings County, Nova Scotia.

These trout were kept 2 months in the aquarium of the Biology Building, Life Sciences Center, Dalhousie University. Two sizes of fiberglass tanks were used in the laboratory: holding tanks, 1.5 × 1.5 × 0.9 m; and acclimation and test tanks, 0.9 × 0.6 × 0.5 m. These aquaria were equipped with continuously flowing dechlorinated fresh water and continuous aeration. The fish were fed commercial fish pellets daily. They were held at temperatures 6–15°C and at pH 6.65–6.90 in the holding tanks before acclimation and testing.

Samples of 10 fish, ranging overall from 96–184 mm and 9.7–62.8 g, were used in each test. A test consisted of exposure of the trout to a regulated pH at either 10 or 20°C until death or until 10 000 min had elapsed. A control group consisted of 10 trout maintained in similar conditions at pH 6.8.

Sulfuric acid and sodium hydroxide were used to regulate the levels of pH at 2.2, 3.2, 3.6, 4.2, 4.8, 5.2, 5.6, 6.8, 9.0, 9.5, 10.0, and 10.8. Apparatus for constant pH control consisted of a continuous-flow dilution system, with a resultant discharge of 66 litres/h of pH-adjusted dechlorinated fresh water. This flow provided a 3-h flushing time for the test tank.

Previously determined volumes of H_2SO_4 or NaOH were metered from an Abbott soluset intravenous constant-delivery reservoir to obtain precise levels of acidity or alkalinity in the constant-head mixing reservoir. The level of pH was monitored frequently both in the mixing reservoir and in the test tank using a Radiometer model 29 pH meter. Variation in pH greater than 0.05 units was of negligible occurrence in any of the tests.

The level of dissolved oxygen was monitored daily by a modified Winkler titrimetric procedure (Strickland and Parsons 1965) and the values were translated to percentage of air saturation (Mortimer 1956). They were kept between 80 and 105%.

The moment of death of each dying fish was observed and the elapsed time noted. The criterion of death was the cessation of opercular movement. Total length and weight at death were recorded and each fish was marked with a numbered tag for reference.

The relation between length and order of death, and between weight and order of death was tested by Spearman rank correlations (Snedecor 1956), for acidic and alkaline media at each temperature and at combined temperatures. The hypotheses tested were that smallest fish died first, and that the largest fish died first.

Results

Percentage mortalities and geometric mean resistance times are summarized in Table 1. Approximate tolerance levels ranged from pH 3.6 to 9.5. Determination of the median lethal levels by probit analysis could not be effected with the data because of the abrupt shift from less than 50% to 100% mortality, resulting in insufficient intermediate levels to permit the calculation of a dose/mortality regression equation. The lower lethal limit occurred within the range pH 3.6–3.2. Visual interpolation of percentage mortalities suggests the lower lethal limit to be pH 3.5. The upper lethal limit occurred within the range pH 9.5–10.0. Interpolation of the percentage mortalities suggests the upper lethal limit to be pH 9.8.

The geometric mean resistance times were similar for both test temperatures at each pH level after acclimation at 15°C.

Results of Spearman rank correlation tests indicated that for brook trout of the length and weight ranges in these tests the order of death was not related to either measure of size.

Discussion

The present study indicated that brook trout have pH lethal limits of 3.5 and 9.8. Only two of the previously reported studies contained a similar systematic approach to the elucidation of the lethal limits of pH to fish. These considered the bluegill, *Lepomis macrochirus*, to have lethal

TABLE 1

Summary of percentage mortality and geometric mean survival time of samples of 10 fingerling brook trout (*Salvelinus fontinalis* (Mitchill)) exposed to ambient pH until death or until 10 000 min had elapsed

pH	Temp.	No.	Mortality, %	Geometric mean time, min
2.2	10	10	100	43
	20	10	100	36
3.2	10	10	100	430
	20	10	100	416
3.6	10	10	20	— ^a
	20	10	10	— ^b
4.2	10	10	0	10 000
	20	10	0	10 000
4.8	10	10	0	10 000
	20	10	0	10 000
5.2	10	10	0	10 000
	20	10	0	10 000
5.6	10	10	0	10 000
	20	10	0	10 000
6.8	10	10	0	10 000
	20	10	0	10 000
9.0	10	10	0	10 000
	20	10	0	10 000
9.5	10	10	0	10 000
	20	10	0	10 000
10.0	10	10	100	1 090
	20	10	100	1 060
10.8	10	10	100	22
	20	10	100	20

^aDeath times 4360 and 5700 min.

^bDeath time 5980 min.

limits of pH 4 and 10.5 (Trama 1954), and the white sucker, *Catostomus commersoni*, to have an acidic lethal limit of pH 4.0 (Beamish 1972). In each instance, acclimation and control tests were accomplished with hard tap water of pH about 7.6.

Proper acclimation procedures and temperature controls at which tests were conducted were mostly lacking in previous studies. Ambient pH regulation of test waters in several studies also appear inadequate. Dahl (1927) used peat acid water diluted with tap water of undetermined quality to regulate pH. His determination of lethal pH at 5.0 for brown-trout fry was measured with a brom cresol purple indicator. Bandt (1936) leached fresh blocks of building concrete to regulate pH. He concluded that pH 9.2 was lethal for perch (*Perca fluviatilis*) and trout, and pH 10.4–10.8 was lethal for roach (*Rutilus rutilus*), pike (*Esox lucius*), carp (*Cyprinus carpio*), and tench (*Tinca tinca*). Eicher (1946) used field observations in Arizona lake waters to conclude

that rainbow trout died at pH 8.8 under certain physiochemical conditions, while they died at pH 10.8 under other conditions.

Results of the present study also indicate, surprisingly perhaps, that there was no more severe gross stress imparted by exposure to greater metabolic demands at 20°C than that observed at 10°C. However, an adverse thermal influence would be expected to occur at temperatures beyond 22–23°C since the ultimate upper lethal temperature of brook trout is 25.3°C (Fry *et al.* 1946).

Although the range of size of brook trout in this study does not span the entire range of size for the species, there is some basis for performing rank correlation since there is a doubling of total length and a ninefold increase in weight among all subjects used in all tests. Intuitively, we think that the lack of correlation between order of death and such size differences implies that the cause of death must be relatively simple and systemic. Subsequent reports will provide the histologic basis for chronic debilitation and death.

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